

METRIX

Online Verification and Digital Certification System for Weighing and Measuring Instruments



A

Project-I Report

Submitted

In Partial Fulfilment

For the award of the Degree of

Bachelor of Technology

(VII Semester)

Problem Statement ID: SIH26036

Supervised By: Guide Name: Mamta Garg Designation: Dean (CSE Dept.)	Submitted By: Vidit J Tiwari (23EJICS178) Tamanna Parveen (23EJICS169) Vishnu Kant (23EJICS182) Section: C Group: 7C2
--	--

Department of Computer Science and Engineering
Jodhpur Institute of Engineering and Technology (Autonomous)
JIET Universe, Jodhpur
Session 2026-27

Certificate

This is to certify that the project entitled "METRIX – Online Verification and Digital Certification System for Weighing and Measuring Instruments" has been carried out by Vidit J Tiwari, Tamanna Parveen and Vishnu Kant, students of Jodhpur Institute of Engineering & Technology, Jodhpur, under my guidance and supervision, in partial fulfilment of the degree of Bachelor of Technology in Computer Science and Engineering of Bikaner Technical University during the academic year 2026-27.

Date:

Place: Jodhpur

Supervisor's Name

Designation

Chapter 2

Literature Survey

2.1 Introduction

The literature survey for this project was conducted by studying international standards publications, research papers, patents and government portals related to legal metrology, digital certification of measuring instruments, and e-governance workflow digitization. The reviewed literature helped in understanding how national and international metrology bodies are moving from paper-based verification records to digital, QR-authenticated certificates, and how such systems are architected for security, interoperability and public trust. Based on these findings, METRIX was designed with dedicated modules for stakeholder and instrument registration, verification application and scheduling, digital inspection and certification, and public QR-based certificate authentication.

2.2 Literature Reviewed

1. OIML Bulletin, “Digital Transparency in Legal Metrology” (2025) – discusses how national metrology institutes, such as Austria's Federal Office of Metrology and Surveying (BEV), have implemented QR-code based digital verification certificate portals (MMEII) that let consumers instantly confirm an instrument's verification validity, and examines the emerging EU Digital Product Passport concept for machine-readable compliance data.
2. “A Secure Software Framework for Measuring Instruments in Legal Metrology” – proposes a digital certificate-based solution for smart meters and fuel dispensers, introducing a new class of digital certificate (“Metrologic Objects Digital Certificate”) approved by Brazil's ICP-Brasil authority, aimed at reducing tampering and improving instrument reliability.
3. PTB (Physikalisch-Technische Bundesanstalt), “Digital Transformation in Legal Metrology: An Approach to a Distributed Architecture for Consolidating Metrological Services and Data” – outlines a modular, REST-based distributed platform implementing a complete digital calibration certificate workflow, designed so new metrological services can be added with minimal rework.
4. “Design and Implementation of an Ontology for Measurement Terminology in Digital Calibration Certificates” (2024) – develops a machine-readable terminology model for digital calibration certificates, verified through semantic query validation, demonstrating how structured, interoperable certificate data can be exchanged across systems and stakeholders.
5. Ministry of Consumer Affairs, Government of India – launch of the ‘e-Maap’ portal (2026), a unified national digital platform for legal metrology services covering registration, licensing, verification and enforcement of weights and measures, announced as part of the Digital India initiative to reduce bureaucratic delay and improve transparency.
6. “Certificateless Digital Signature Technology for e-Governance Solutions” – presents an architecture for issuing digitally signed government certificates without relying on a centrally

managed certificate authority, using cryptographic key pairs bound to the issuing officer's identity to guarantee that a certificate's content and issuing authority can be verified and that any post-issuance tampering is detectable, a model relevant to securing METRIX's officer-issued digital verification certificates.

7. "Anti-Counterfeit Framework of Electronic Certificate Based on QR Code and Seal Watermark," Multimedia Tools and Applications (2024) – proposes a triple-verification scheme that embeds watermark data inside a certificate's QR code and official seal image, cross-matching the printed, encoded and watermarked information to detect forged or reprinted certificates, a security model directly applicable to METRIX's public QR-based certificate authentication module.
8. "Transforming Public Service Delivery: A Comprehensive Review of Digitization Initiatives," Sustainability, MDPI (2024) – reviews e-governance digitization programmes worldwide and finds that citizen-centric design, role-based service delivery and measurable efficiency gains are the strongest predictors of successful adoption, reinforcing METRIX's multi-stakeholder, role-based approach to Legal Metrology service delivery.

2.3 Key Findings from the Literature

The literature survey led to the following key observations, which directly shaped the design of METRIX:

- QR-code based digital verification certificates significantly improve transparency, allowing consumers and regulators to instantly authenticate an instrument's status without physical paperwork.
- Digital-certificate mechanisms bound to individual instruments can reduce tampering and fraud on regulated equipment such as fuel dispensers and weighing scales.
- Modular, service-oriented architectures allow verification and certification platforms to be extended with new services over time without redesigning the whole system.
- Standardised, machine-readable certificate formats improve interoperability of verification data across departments, test centres and public-facing verification tools.
- National initiatives such as India's e-Maap portal confirm strong government commitment toward digitizing registration, licensing, verification and enforcement on a single platform — validating the direction of the proposed project.

2.4 Research Gap Identified

Despite the growing body of work on digital metrology and certification, the reviewed literature shows that most existing efforts either target calibration certificates for laboratory/industrial instruments, or address a single instrument category such as smart meters or fuel dispensers, rather than a unified, end-to-end platform spanning stakeholder registration, application submission, scheduling,

inspection, certificate issuance and validity tracking for the full range of weighing and measuring instruments used in everyday commercial transactions.

Further, India-specific initiatives such as e-Maap are still at an early, centrally-operated portal stage, with no openly documented, reusable reference architecture that state-level Legal Metrology Departments, Government Approved Test Centres (GATCs), or academic teams can study or adapt. There is also limited literature that combines public QR-based certificate authentication with role-based officer scheduling and automated expiry-reminder workflows within a single lightweight system — this is the specific gap that METRIX aims to address.

2.5 Outcome of the Study

Based on the literature survey and the gaps identified, the outcome of the study is a platform architecture built around five core pillars:

- Stakeholder & Instrument Registration
- Verification / Re-verification Application & Scheduling
- Digital Inspection & Certification (QR-enabled)
- Public Certificate Authentication
- Validity Tracking, Notification & Reporting

Together, these pillars form the foundation of the METRIX platform, ensuring that an instrument moves from application submission to a verifiable, QR-authenticated digital certificate, with role-based oversight and transparency at every stage.

Chapter 3

Requirement Specification & Methodology

3.1 Software Development Life Cycle (SDLC) Model – Agile Methodology

METRIX is being developed using the Agile methodology, an iterative and incremental approach to software development in which requirements and solutions evolve through collaboration between the development team and stakeholders, and the system is built in small, workable increments called sprints rather than as one large, monolithic release. Agile was selected over sequential models such as the traditional Waterfall model because the Waterfall model requires requirements to be frozen at the outset and does not accommodate the kind of iterative feedback and refinement this project needs, whereas METRIX involves several distinct stakeholder types — instrument owners/businesses, Legal Metrology Officers (LMOs), Government Approved Test Centres (GATCs), administrators, and members of the public verifying certificates — each with workflows and expectations that are best understood and refined through repeated review rather than specified completely in advance.

In the context of this project, Agile allows functionally independent modules — such as stakeholder registration, verification-application handling, inspection scheduling, digital certificate generation, and expiry notification — to be designed, developed, tested, and demonstrated as separate, self-contained increments. Each increment is reviewed against the SIH26036 problem statement and refined based on feedback from the project guide and simulated end users before the next module is taken up, rather than waiting until the entire system is complete to discover design gaps or usability issues. This reduces the risk of late-stage rework, allows the highest-priority modules (such as application submission and certificate issuance) to be stabilized early, and gives the team a continuously working, demonstrable prototype at the end of every sprint instead of a single deliverable only at the very end of the development period.

The Agile process followed for METRIX consists of four broad, overlapping phases, each treated as one or more short sprints: (i) requirement analysis and system design, during which the Legal Metrology verification workflow was studied and translated into functional and non-functional requirements, system architecture, and database schema; (ii) core system development, covering authentication, stakeholder and instrument management, and application submission and tracking; (iii) verification, scheduling and certification, covering officer allocation, inspection recording, and QR-enabled digital certificate generation; and (iv) dashboarding, notification and integration, covering role-specific dashboards, expiry alerts, reporting, and end-to-end testing of the integrated system. At the end of each phase, the working increment is reviewed, defects are logged and fixed, and the backlog for the following phase is adjusted accordingly — which is the essence of the Agile, sprint-based methodology applied to this project.

3.2 Project Workflow

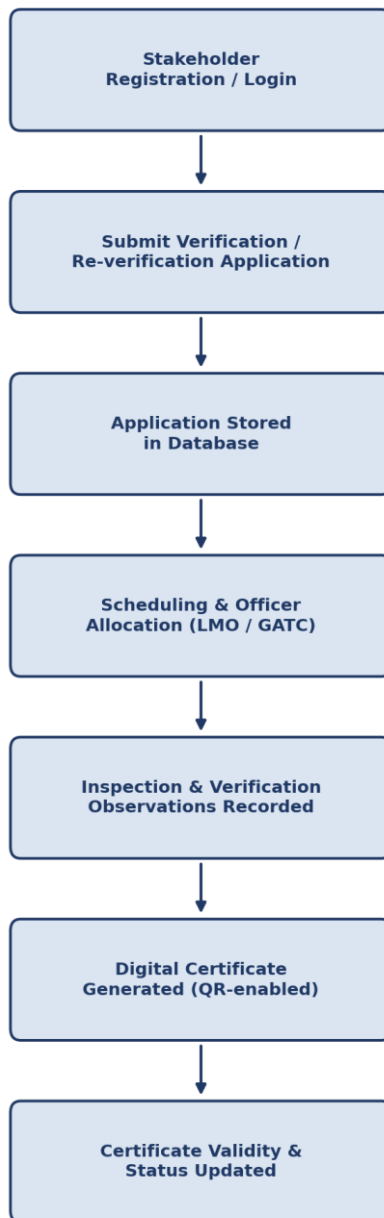


Figure 3.1: Project Workflow of METRIX

The core workflow of the platform, from a stakeholder submitting a verification application to certificate issuance, follows these steps:

1. Stakeholder Registration / Login
2. Submission of Verification / Re-verification Application
3. Application Stored in Database
4. Scheduling and Officer Allocation (LMO / GATC)
5. Inspection and Verification Observations Recorded

6. Digital Certificate Generated (QR-enabled)
7. Certificate Validity and Status Updated

3.3 Main Modules of the System

The system is organized into the following nine main modules:

1. User Authentication & Role-Based Access Module
2. Stakeholder & Instrument Management Module
3. Verification / Re-verification Application Module
4. Scheduling & Officer Allocation Module
5. Verification & Inspection Module
6. Digital Certificate & QR Generation Module
7. Public QR Certificate Authentication Module
8. Notification & Expiry Tracking Module
9. Dashboard, Reporting & Database Module

3.4 Use Case Diagram

A use case diagram is a UML diagram that represents the interaction between users/actors and a system. The diagram below shows the five actors of METRIX — Instrument Owner/Business, Legal Metrology Officer (LMO), GATC (Test Centre), Administrator, and Public/Consumer — and the functionalities each of them can access.

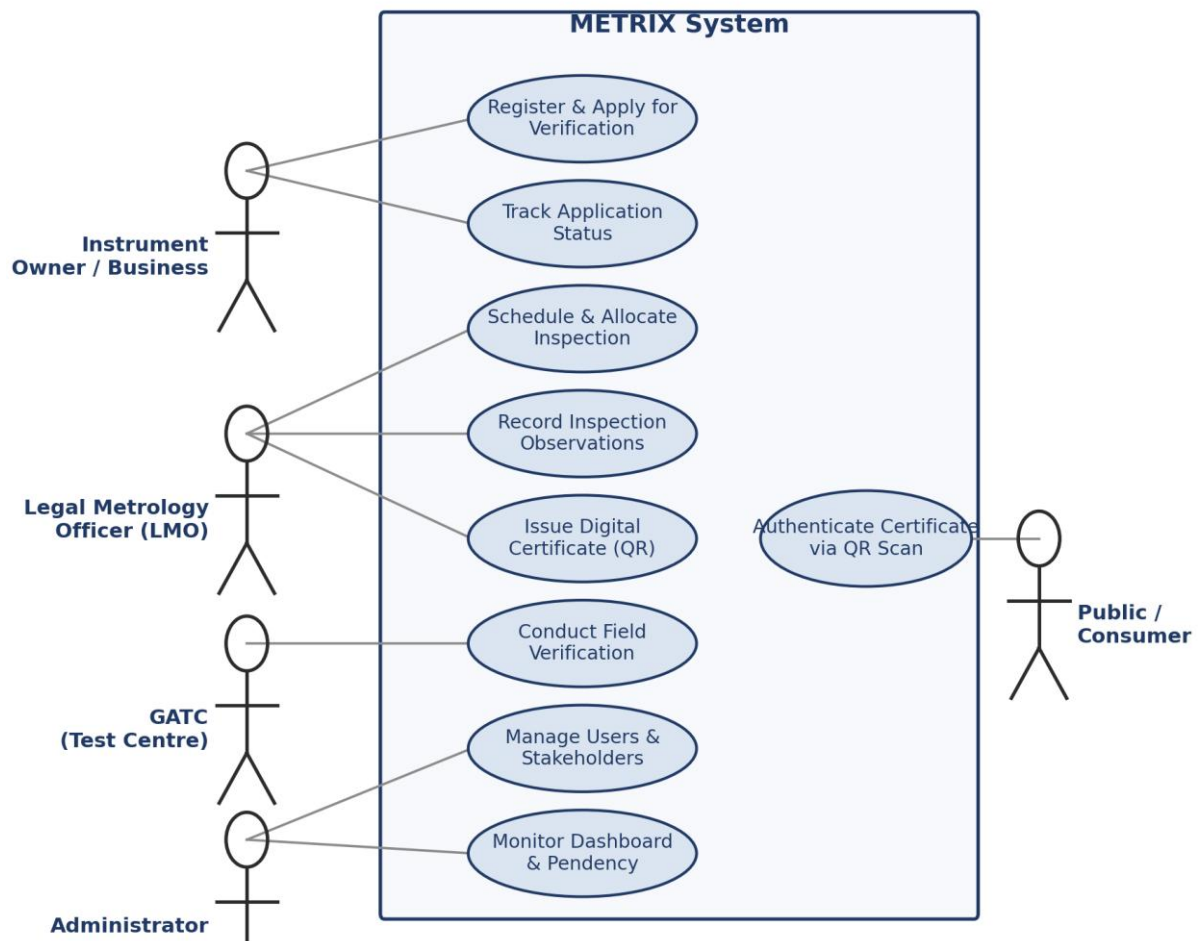


Figure 3.2: Use Case Diagram of METRIX

Actor-wise Use Cases

- Instrument Owner / Business – Register/Login, Submit Verification Application, Track Application Status
- Legal Metrology Officer (LMO) – Review Applications, Schedule Verification, Record Inspection Observations, Issue Digital Certificate
- GATC (Test Centre) – Receive Allocated Inspections, Conduct Verification, Record Results
- Administrator – Manage Users & Stakeholders, Monitor Dashboard & Pendency, Generate Reports
- Public / Consumer – Authenticate Certificate via QR Scan, View Verification History